OVERTURE MAPS FOUNDATION

A

Places Attributes Conflation

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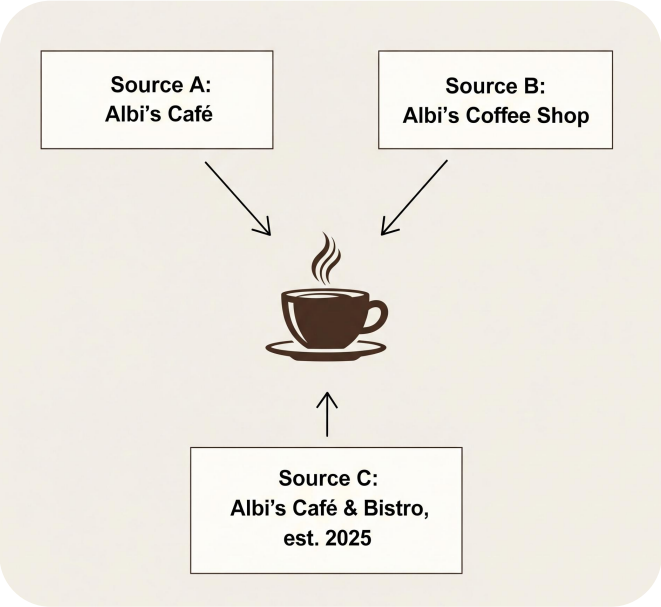
Too Many Truths



One real-world place,
multiple data sources.

All with slightly different information.

Our problem:
How do we create a single,
reliable record?

Jacob’s Approach - Overview I (Plan)

- Which can be resolved before ML or external verification?
- Start from the data itself
- Understand what conflict really means, what kinds of conflicts exist
- Goal?
  - Get insights for feature engineering, discover hidden features
  - Building rule-based algorithms
  - Documenting exactly where rules end and ML begins
- Maybe answer questions like…
  - “Which provider gets what right?”
  - “Confidence is a row level attribute and two-state flag, with small gaps, but can it help predict all or certain attributes?”
  - “Can we take both attributes and construct a better one?”’
  - Hopefully more!

Jacob’s Approach - Phones I (Null Normalization)

Column	Meaning	Base Count	Alt Count
`total_rows`	Total candidate rows evaluated per source	2000	2000
`sql_null`	True SQL NULL values	4	109
`bracket_null`	Exact string `[null]`	288	0
`bracket_empty_string`	Exact string `[""]`	8	10
`bracket_null_caps`	Exact string `["NULL"]`	1	0
`bracket_null_lower`	Exact string `["null"]`	0	0
`contains_null_text`	Any non-null string containing the substring `null`	289	0

Null normalization resolves 206 false conflicts

Jacob’s Approach - Phones II (Staged Pipeline)

- Apply progressively stronger normalization rules
- Measure how apparent conflict drops at each stage
- Each stage is monotonic
- Percentages off of 1,680 usable rows (both sides have at least one digit)
- The remaining 320 rows: 100 have no phone on either side, and 220 have a phone on only one side (201 alt-only, 19 base-only)

Stage	Rule	Conflict Count	Conflict Rate	Improvement vs Prior
S0	Null normalization (preamble)	—	—	—
S1	Raw string compare	1330	79.17%	—
S2	Strip non-digits; exact match	1261	75.06%	5.19%
S3	US: +1 leading digit vs 10-digit	819	48.75%	35.05%
S4	Generic 1-digit CC drop	765	45.54%	6.59%
S5	Trunk 0 vs 2-digit CC	554	32.98%	27.58%
S6	Trunk 0 vs 3-digit CC	478	28.45%	13.72%
S7	2-digit CC vs national (no trunk)	425	25.30%	11.09%
S8	3-digit CC vs national (no trunk)	417	24.82%	1.88%
S9	CC2+0+national vs CC2+national	402	23.93%	3.60%

Jacob’s Approach - Phones IV (Interpretation)

- Raw string comparison overstates phone conflict by **3.3x**
- In other words, conflict rate of **79.17% to 23.93% (a 69.77% improvement)**
  - the true pre-normalization conflict rate is 1536 (91.4% of the 1680 usable rows), making the actual overstatement factor approximately **3.8x** rather than 3.3x.
- The remaining ~24% (402 pairs) represent genuinely different phone numbers. These are true semantic conflicts requiring reconciliation logic
  - Manually verified (ONE or two edge cases)
  - Abstention
    - Neither rule-based nor ML approach can determine correctness without external verification

Jacob’s Approach - Names

Bucket	Rows	% of Total	Meaning
`full_agreement`	625	31.3%	Entire JSON matches — no conflict at all
`primary_agrees_json_differs`	292	14.6%	Primary names match; JSON envelope differs (empty fields)
`primary_conflict`	1083	54.2%	Primary names disagree — this is what we classify below

- Raw string comparison,
- No normalization yet

- alt: `{"primary":"Edward Jones"}`
- base: `{"primary":"Edward Jones","common":{},"rules":[]}`

Jacob’s Approach - Names - Two-tier Classification

- Decision tree
  - Casing only? *(Tier 1)*
  - Normalization equivalent? *(Tier 1)*
    - Staged monotonic pipeline
    - What kind? *(Tier 2)*
  - Subset? *(Tier 1)*
    - What kind? *(Tier 2)*
  - Genuinely different. Catch-all. *(Tier 1)*
- Gets one label and stops

Jacob’s Approach - Names - Tier 1

Tier 1 Label	Meaning	Reconciliation Action
`casing_only`	Identical after lowercasing	Auto-resolve; formatting pref
`normalization_equivalent`	Same name after punctuation/spacing/diacritic/conjunction/spelling/script-form normalization	Auto-resolve; formatting pref
`subset`	One name is meaningfully contained in the other (brand vs brand+branch, name vs name+gloss)	Policy decision: prefer core name vs full listing
`genuinely_different`	Different names for the same place, or potential bad match	Requires human review or abstention

- Over 82% of name conflicts (casing + normalization + subset) have a deterministic or policy-based resolution path. Only 17.3% are genuinely different names requiring human review.

The distribution across 1083 conflict rows:

Tier 1 Label	Count	% of Conflicts
`subset`	600	55.4%
`normalization_equivalent`	238	22.0%
`genuinely_different`	187	17.3%
`casing_only`	58	5.4%

Jacob’s Approach - Names (T1) - 1. Casing Only

- 1. CASING:

Straightforward (prefer title case, and lowercase articles/prepositions):

- CVS Pharmacy v CVS pharmacy
- Honda Of Jonesboro v Honda of Jonesboro
- Subway v SUBWAY
- davaindia GENERIC PHARMACY vs Davaindia Generic Pharmacy
- Sklep rybny SIEJA vs Sklep Rybny Sieja

Language-specific casing conventions:

- Fleurs d'Alain v Fleurs D'Alain
- Osteria del Tempo Perso v Osteria Del Tempo Perso
- セルフ写真館BLANC v セルフ写真館Blanc

Owner’s stylistic choice: (How do we know when capitalization is an abbreviation versus a stylistic choice?)

- OXXO vs Oxxo, bp vs BP
- RAQSA Radiadores vs Raqsa Radiadores
- Shell v SHELL, ROSSMAN vs Rossmann
- HDFC Bank ATM vs Hdfc Bank Atm
- 1 edge case: Conad v CONAD

Less straightforward (intentional camelCase):

- PuroClean v Puroclean
- IndianOil vs Indianoil
- ecoATM vs Ecoatm
- 1 edge case: Cocktailz Fine Wine and Spirits v CockTailz Fine Wine and Spirits

Edge cases:

- THE YARD, LLC v The Yard LLC
- Denns BioMarkt v Denn’s Biomarkt

Jacob’s App. - Names (T1) - 2. Normalization Eq.

#	Check	What it catches	Example
1	`norm_compare` match	Case, accents, punctuation differences	`Origen's Bistrô` → `origens bistro` ← `origensbistro`
2	Space-stripped match	Compound split/join	`Photo Color Lab` → `photocolorlab` ← `PhotoColorLab`
3	Conjunction-normalized	`&` / `and` / `et` / `und` / `y` / `e` unified	`Acqua e Sapone` → `acqua and sapone` ← `Acqua & Sapone`
4	Conjunction + space-stripped	Conjunction + compound boundary	`Crepes&Coffee` → `crepesandcoffee` ← `Crepes & Coffee`
5	Spelling-normalized	British/American variants	`Defence Centre` → `defense center` ← `Defense Center`
6	Word-reorder (sorted tokens)	Same words in different order	`Colombo Cristoforo` → `colombo cristoforo` ← `Cristoforo Colombo`
7	Katakana→hiragana	Japanese script-form variants (see note below)	`コインランドリーハナコ` → `こいんらんどりーはなこ` ← `コインランドリーはなこ`
8	Katakana→hiragana + space-stripped	Script-form + Japanese compound boundary	Same as 7 but also collapses spacing differences
9	Levenshtein typo (see note below)	1–2 character substitutions/insertions	`คลีนิคบ้านหมอ` ↔ `คลีนิกบ้านหมอ` (Thai spelling variant)

Jacob’s App. - Names (T1) - 2. Normalization Eq. Counts

For `normalization_equivalent` **rows:**

Subtag	Meaning
`punctuation`	Dash/dot/apostrophe/quote/nakaguro (・) differences
`spacing`	Compound split/join (PhotoColorLab ↔ Photo Color Lab)
`diacritic`	Accent differences (Bistrô ↔ Bistro, Creatività ↔ Creativita)
`conjunction`	&/and/et/und/y/e interchange
`spelling`	British/American variants (centre ↔ center)
`word_reorder`	Same tokens in different order
`script_form`	Katakana ↔ hiragana (ハナコ ↔ はなこ), small kana variants
`typo`	Levenshtein distance ≤ 2 (only when no other subtag explains it)

Normalization subtags (238 rows;

Subtag	Count	% of Norm. Rows
`punctuation`	85	35.7%
`spacing`	76	31.9%
`typo`	37	15.5%
`word_reorder`	26	10.9%
`diacritic`	19	8.0%
`conjunction`	15	6.3%
`script_form`	1	0.4%
`spelling`	1	0.4%

Counts up everything, not classify

Jacob’s Approach - Norm. Equiv. - T2 Conjunction

CONJUNCTION

- Normalizer works great.
- Howarth Timber and Building Supplies v Howarth Timber & Building Supplies

- Acqua e Sapone vs Acqua & Sapone

- Open question
  - Should the golden record prefer & universally
  - or preserve the original language's conjunction (e in Italian, y in Spanish, und in German)?

- One approach: use the language of the name attribute rather than GPS coordinates.

- A French-language name in Texas should still use et if that is what the business uses.
- GPS coordinates can help as a tiebreaker when the name language is ambiguous, but should not override name-level language signals.

Note: conjunction insertion/deletion

- (`Bäckerei Konditorei` vs `Bäckerei & Konditorei`) is not handled — only substitution is normalized.

Jacob’s Approach - Norm. Equiv. - T2 Diacritic

DIACRITIC

Clear cases:

- Imobiliária Alegro vs Imobiliaria Alegro
- Café de l'Harmonie vs Cafe de l'Harmonie

But what about American businesses want to get fancy?

- Wildberry Pancakes and Café vs Wildberry Pancakes and Cafe

Accent should be preserved in languages that use it

Prefer the form with more linguistic information

Another approach:

- As with conjunctions, the name's own language is a stronger signal than GPS
- If the name is English with a borrowed word, the owner may have chosen either form.

Potential edge case: If an Italian café uses English branding with an accented Café, that may be intentional stylization that cannot be resolved without owner input.

Jacob’s Approach - Norm. Equiv. - T2 Punctuation

Dashes as separators: Very common, easily normalized

- Farmers Insurance - David Hiney vs Farmers Insurance David Hiney
- Maricopa County Sheriff's Office - District III Substation vs Maricopa County Sheriff's Office District III Substation.

Standard: drop separator dashes. But dashes within compound words may reflect owner's choice:

- Grill-Ecke vs Grillecke (German compound).
- A1 Dental vs A-1 Dental.
- Save A Lot vs Save-A-Lot.

Abbreviation dots: Prefer periods

- Dana Stampi s.r.l. vs Dana Stampi SRL,
- D.P.T. vs DPT.
- Lekáreň Dr.Max v Lekáreň Dr. Max
- Scuola Materna Statale 44 Circolo Didattico G.L. Radice v Scuola Materna Statale 44 Circolo Didattico G. L. Radice
- União Vila Sá F.C. v União Vila Sá F. C.
- Baghdad Food Mart v Baghdad foodmart

Apostrophes: Preserve, carries meaning.

- Aherne's vs Ahernes
- L'ynara Brautmode vs Lynara Brautmode
- Dunkin' Donuts vs Dunkin Donuts
- Edge case: L’occitane au Brésil v L’Occitane au Brésil (French elision)
- Murkel's Nähwelt vs Murkelsnähwelt

Quotation marks: Not enough examples, no rule yet

- Centro Aperto Polivalente per minori "LOL" vs Centro Aperto Polivalente per Minori Lol

Jacob’s Approach - Norm. Equiv. - T2 Punctuation

Plus signs and special charts: Strip

- PostalAnnex+ vs PostalAnnex
- Brothers Mechanical Services vs Brothers Mechanical Services®
- Edge case: Gas vs Gas+

Branding elements, but flag as owner-decided

Exclamation mark: Strip or keep? (not enough examples)

- Maloserá vs Maloserá!

Likely owner’s stylistic choice

Japanese nakaguro (・)

- ビジネスホテル・キャッスル vs ビジネスホテルキャッスル
- Different than putting Katakana in parentheses, this is to separate loanwords
- Most likely keep

Extra conjunction:

- Müller & Egerer Bäckerei Konditorei vs Müller & Egerer Bäckerei & Konditorei

Spacing, Typo

- Open question: could a typo dictionary or fuzzy-match library improve resolution? Levenshtein catches 1–2 character differences, but has no concept of common misspellings.

Jacob’s Approach - Norm. Equiv. - T2 Punctuation

Edge cases:

- Gimnasio R&C vs Gimnasio RYC (Will fix)
- Small kana variants (ミカモラィディングクラブ vs ミカモライディングクラブ) and fullwidth Latin (ヘアーサロンa-cubu vs ヘアーサロンa-cubu) are both correctly resolved by the existing pipeline (Levenshtein and NFKD normalization respectively)

Jacob’s Approach - Names - T1

3. SUBSET - If one name is contained inside the other, the row is `subset`.

4. GENUINELY DIFFERENT - Catch all

- Example: `ก๋วยเตี๋ยวปลาในตำนาน` (Legendary fish noodles) vs `ก๋วยเตี๋ยวปลาสด` (Fresh fish
- noodles) — different words, different business names

Jacob’s Approach - Names (T2) - Subset

For `subset` rows:

Subtag	Meaning
`branch_suffix`	Brand + store/location name (very common in JP/TH data)
`parenthetical`	Name + parenthetical reading/disambiguation
`biz_suffix`	Legal suffix added/removed (LLC, GmbH, SRL, Inc, ...)
`seo_junk`	SEO keywords in the longer name (hours, directions, near me)
`facility_suffix`	Bus stop, ATM, substation appended
`descriptor`	Additional business description (catch-all)

Subset subtags (600 rows):

Subtag	Count	% of Subset Rows
`descriptor`	442	73.7%
`biz_suffix`	52	8.7%
`branch_suffix`	46	7.7%
`parenthetical`	36	6.0%
`facility_suffix`	29	4.8%
`seo_junk`	1	0.2%

Jacob’s Approach - Names - T1 Subset - T2 Tags

Business Suffix

- LLC, Inc, GmbH, SRL, s.r.l., etc., is partly legal identity, partly owner preference
- Stripping suffixes for comparison is useful
- But golden record should preserve or omit them based on policy

Proposal: prefer to keep

Facility Suffix

- Japanese bus stops append “バス停”:
  - 粟生団地 vs 粟生団地バス停, 東名吉田 vs 東名吉田バス停.
- English ATMs: Barclays vs Barclays ATM.

Proposal: prefer both.

Branch Suffix

- Consistent pattern: one side has the brand name, the other has brand + branch/store location.
- Japanese
  - JINS vs JINS イオンモール宮崎店 (brand vs Aeon Mall Miyazaki branch)
  - セブンイレブン vs セブンイレブン 波崎土合南店 (7-Eleven vs 7-Eleven Hasaki branch)
  - Hardcoded: 店, 支店, 出張所, 営業所, イオン, モール, マルイ (store, branch, office, AEON, mall, Marui, etc.)
- English hardcoded: Branch, Store, Center, Plaza, Outlet, Location, Mall
- Thai hardcoded: สาขา, ถนน, ซอย, ต., อ., จ. (branch, road, alley, subdistrict, district, province)

Proposal: Core name (shorter), but do international dense areas prefer full branch listing (more specific, better for disambiguation?)

Jacob’s Approach - Names - T1 Subset - T2 Tags

Parenthetical

- Thai listings frequently include an English rendering/local nickname in parentheses as an aid.
  - Cultural convention in Thai directory data or SEO noise?
  - However, some are location

1. วิทยาลัยเทคนิคพระนครศรีอยุธยา vs วิทยาลัยเทคนิคพระนครศรีอยุธยา (Phra Na Khon Sri Ayutthaya Technical College)
2. สวนอาหารบ้านตะวันแดง vs สวนอาหารบ้านตะวันแดง (Baan Tawundang Resturant)
3. สะพานนครพิงค์ vs สะพานนครพิงค์ (Nakhonping Bridge)

Japanese — kana reading or tagline: Japanese uses parentheses for furigana (pronunciation guide) or descriptive taglines.

- 虎侍我炎（とらじがえん）【筑後吉井紅豚餃子】 vs 虎侍我炎 — kanji name + hiragana reading + product tagline
  - Torajigaen (Torajigaen) [Chikugo Yoshii Red Pork Gyoza])
  - 虎侍我炎（とらじがえん）
  - Torajigaen (Torajigaen)

- Moana vs Moana (モアナ) — English brand + katakana reading
- みらい平駅 vs みらい平駅 (Miraidaira Sta.) — station name + romanized abbreviation

Jacob’s Approach - Names - T1 Subset - T2 Tags

Other languages are different - simpler.

Parenthetical content tends to be location context or acronym expansion.

- Vitality Integrated Programs vs Vitality Integrated Programs (VIP) — name + acronym
- The Coffee Corner (Hermitage mall) vs The Coffee Corner
- Centre RNPC Avranches vs Centre RNPC AVRANCHES (Quentin sur le homme)
- Makmal Kesihatan Awam (MKA) vs Makmal Kesihatan Awam
- Sídliště Libuš (bus) vs Sídliště Libuš

Proposal:

- For CJK/Thai names, the parenthetical content is preserved — Thai transliterations and Japanese kana readings are essential for disambiguation, not noise.
  - But could better dicts or an SLM learn to get rid of locations?
- For English names, the parenthetical is typically location context or acronym expansion; prefer the form without parentheses
  - Option: preserve the content as metadata where the schema supports alternate names.

Jacob’s Approach - Names - T1 Subset - T2 Tags

DESCRIPTOR

- Largest subset category (gets hard)
- one side adds business description, department names, service lists, or location qualifiers that the other side omits
- エコモベーカリーヨコハマモトマチ vs エコモベーカリー (Ecomo Bakery Yokohama Motomachi vs Ecomo Bakery)
- Clarkson Eyecare Florida vs Clarkson Eyecare (location qualifier)
- โรงแรมพร3 #ขอนแก่น vs โรงแรมพร 3 (hotel + hashtag city name vs just hotel)

In almost every Latin cases the shorter form is the core business name and the longer form is a listing-specific elaboration.

- Check on Thai/Japanese conventions

Standard: prefer the shorter core name. However, this is a policy decision — the owner may prefer the more specific form.

SLM could help.

Prefer full-width spaces.

Jacob’s Approach - Names - T1 Subset - T2 Tags

- hotel, motel, inn, hostel, resort, lodge, bar, pub, cafe, café, bistro, restaurant, grill, pizzeria, trattoria, brasserie, tavern, cantina, spa, salon, studio, clinic, pharmacy, apotheke, hospital, medical, dental, supermarket, market, mart, grocery, store, shop, boutique, gallery, museum, theater, theatre, cinema, school, academy, university, college, institute, church, mosque, temple, cathedral, gym, fitness, arena, stadium, bank, insurance, agency, sportsbar, brewpub, coffeehouse
- ATM, branch, outlet, location, mall, plaza, center, centre

- Latin, drop location
- Prefer punctuation
- Prefer business suffixes
- Prefer longer Japanese/Thai listings, prefer Japanese bus stops

Else, pick shorter name.

Jacob’s Approach - Name Reconciliation - Genuinely Diff

- No normalization can reconcile

Different brand names for the same entity

- Citibanamex 30 Av. Playa Del Carmen vs Citibank — the Mexican subsidiary name vs the global brand
- Concord Foot & Ankle Center vs Concord Feet — same place, different naming conventions.

Different locations entirely:

- Sport Clips Haircuts of Panama City - Cahall's Deli Plaza vs Sport Clips Haircuts of Lynn Haven

Same entity, different naming conventions:

- Barclays ATM vs Barclays Bank Cash Machine
- Mishicot Fire Dept vs Mishicot Volunteer Fire Depart
- the same place described differently by two providers.

Shared prefix with different qualifiers:

- Profile Publishing Location 1 vs Profile Publishing Location Name - Suite
- This is the key distinction from subset, where one name is fully contained in the other.

Jacob’s Approach - Name Reconciliation - Genuinely Diff

Japanese — same brand, different branch encoding

- -ヤマト運輸 YamatoTransport vs ヤマト運輸 伊達センター
  - One side appends an English transliteration, the other appends a Japanese branch name
- ゴルフパートナーpga大宮店 vs ゴルフパートナー PGATOURSUPERSTORE大宮店
  - same store, but one abbreviates the brand and the other uses the full name.

Borderline cases that could be other labels

- สำนักงานอธิการบดี ม.รามคำแหง vs สำนักงานอธิการบดี มหาวิทยาลัยรามคำแหง — ม. is a standard Thai abbreviation for มหาวิทยาลัย (university)
  - Lookup table?
- B&B Hôtels vs B&B Hôtel Lyon Ouest Tassin — brand vs brand + location. This looks like a subset but the brand name itself differs, so substring containment fails.
- 元妙古觀 vs 元妙古观 Yuanmiao Temple — traditional vs simplified Chinese characters plus an English transliteration

Jacob’s Approach - Golden Dataset - Prefer Squished

- Japanese conventionally writes without spaces between words
- Richard Lavigna Dpm v Richard Lavigna, DPM
- Dance4Life v Dance 4 Life, SUMA LIFT v SUMALIFT
- First Choice Haircutters v First Choice Hair Cutters
- Crepes&Coffee v Crepes & Coffee, Space N K v Space NK
- Early County Courthouse v Early County Court House
- Angel's Ayur Spa v Angel’s Ayurspa
- I Dine Chettinad Virundhu v Idine Chettinad Virundhu
- EdShed, PhotoColorLab, BeautySpace, YohCo., SignWriter, TriCity Golf Course, VaVaVooom, MakiMoi, ZonaSuples, PezVaca, HemBurger

- EDGE CASE: DelMoro Supermarket v Del Moro Supermarket
- EDGE CASE: TerasCorner v Teras Corner, NovaCordis v Nova Cordis

- pickleball 406 vs pickleball406

Edge cases:

- Thai (more nuanced) - uses spaces between phrases but not between words within a phrase
- L’occitane au Brésil v L’Occitane au Brésil **(French elision)**
- Murkel‘s Nähwelt v Murkelsnähwelt **(German compound words)**
- Ace Marine GmbH v AceMarine GmbH
- Brierwood Child Care Center v Brierwood Childcare Center
- Treebo G K Residency v Treebo GK Residency
- Moshi CoffeeBar v MoshiCoffeeBar
- A Credilar v Acredilar
- Hotel&Bugalows El Mirador v Hotel & Bugalows El Mirador

Stage	Rule	Conflict Count	Conflict Rate	Improvement vs Prior
S0	Raw primary name comparison	1083	54.15%	—
S1	Casing (lowercase)	1025	51.25%	5.36%
S2	Unicode (fullwidth + diacritics)	1001	50.05%	2.34%
S3	Punctuation stripped	935	46.75%	6.59%
S4	Spaces stripped	867	43.35%	7.27%
S5	Conjunctions unified	854	42.70%	1.50%
S6	Spelling normalized	853	42.65%	0.12%
S7	Katakana→hiragana	852	42.60%	0.12%
S8	Word reorder (sorted)	826	41.30%	3.05%
S9	Typo detection (Levenshtein ≤ 2)	768	38.40%	7.02%
Subset	One name contained in the other	176	8.80%	77.08%

Jacob’s Approach - Staged Normalization Pipeline

Percentages are based on 2000 total rows. Of the 1083 original primary-name conflicts:

Disposition	Rows	% of Conflicts	Meaning
Resolved by normalization	315	29.1%	Same name after formatting noise is stripped
Subset (policy decision)	592	54.7%	One name is contained in the other
Genuinely different	176	16.3%	Irreducible conflict requiring human review

Jacob’s Approach - Staged Normaliz. / Golden Dataset

- Raw string comparison overstates name conflict by approximately 6x. After applying all normalization stages and subset detection, apparent conflict drops from 54.2% to 8.8%.
- Subset detection accounts for majority of remaining conflicts
    - Reconciliation requires policy decision

Hard coded scoring system

If all else fails to detect, prefer the shorter name.

If shorter name is <6 characters or a single word, prefer the longer one

SLM Benefits? Combining / detecting

- When both sides or the winning side is imperfect
- When taking the form with more linguistic information (diracritics), we might lose good formatting/information in other name
    - Auto-école De L'etoile vs Auto-Ecole de l'Etoile
- One listing may have the name, location and noise/SEO/irrelevance, the other just the name
- Or one side might just have the more correct name overall but a small typo or something

Jacob’s Approach - Staged Normaliz. / Golden Dataset

Edge cases

- Tumi vs TUMI Champs-Elysees — Champs-Elysees is a location, should prefer shorter
- Walmart vs Walmart Azle — Azle is a location, should prefer shorter
  - Walmart SuperCenter, Walmart Neighbourhood Market
- BP vs BP America — America is not a useful qualifier (external lookup)
- Schlotzsky's vs Schlotzsky's (Colony Way - Madison, MS) — location in parentheses
- Sherwin-Williams Paint Store vs Sherwin-Williams

Outcome	Rows	% of Total	Meaning
Agreement (no conflict)	917	45.9%	Both sides identical — golden name is obvious
Rule-selected with confidence	~850	~42.5%	Rules produce a defensible selection
Rule-selected with caveats	~50	~2.5%	Selection is plausible but edge cases exist
Abstain (human review)	~180	~9.0%	Rules cannot determine the better name

	Phones	Names
Total rows	2000	2000
Both present	1680 (84%)	2000 (100%)
One-sided (trivial)	220 (11%)	0
Both null	100 (5%)	0
Conflicts (of comparable rows)	1330 / 1680 (79.2%)	1083 / 2000 (54.2%)
Resolved by normalization	928 (69.8%)	315 (29.1%)
Resolved by subset	—	592 (54.7%)
Remaining true conflicts	402 (20.1% of all rows)	176 (8.8% of all rows)

Jacob’s Approach - Note

Accent stripping must distinguish Latin diacritics from script-essential marks

- Unicode represents Latin characters as a base letter plus a combining mark
  - (e.g. é = e + combining acute accent)

What about marks that are linguistically meaningful in other scripts?

- The fix is to only strip combining marks in the Latin Combining Diacritical Marks block (U+0300..U+036F), leaving Japanese, Thai, Korean, Arabic, and other script-specific marks intact.

In Japanese, the dakuten is a combining mark that voices consonants

- two small diagonal strokes placed in the upper-right corner of hiragana or katakana
- stripping it turns バ (ba) into ハ (ha), changing the character's pronunciation and meaning entirely

In Thai, tone marks (่ ้ ๊ ๋) are combining marks that determine which of five tones a syllable carries

- stripping them changes word meaning (e.g., ห้วย "stream" loses its tone mark and becomes ambiguous)

More examples, but 5% total of 2000

Note#2: *CJK text commonly uses the fullwidth ideographic space (U+3000) instead of the ASCII space (U+0020). These render identically in most fonts

David’s approach

Idea

Replace “mayhem” with an auditable, weighted policy system that reduces over-merging and increases record-level accuracy.

Methodology & Architecture for “v2”

Two-Layer System: attribute layer - extracts features like phones, category, website, etc.

record layer - aggregates those votes into a final “keep or delete” decision.

Key Benefit

This allows us to tune individual attribute weights without retraining the entire pipeline.

Benchmarks

Before deciding on a record I had to run some benchmarks

coverage = count of predictions made/ total number of samples in dataset

the f1 is the harmonic mean where
f1 = 2 * (precision*recall) / (precision+recall)

precision being true positive/(true positive+false positive) and recall = tp / (tp +fn)

The key takeaway from the results here is that name and addresses were major weaknesses…

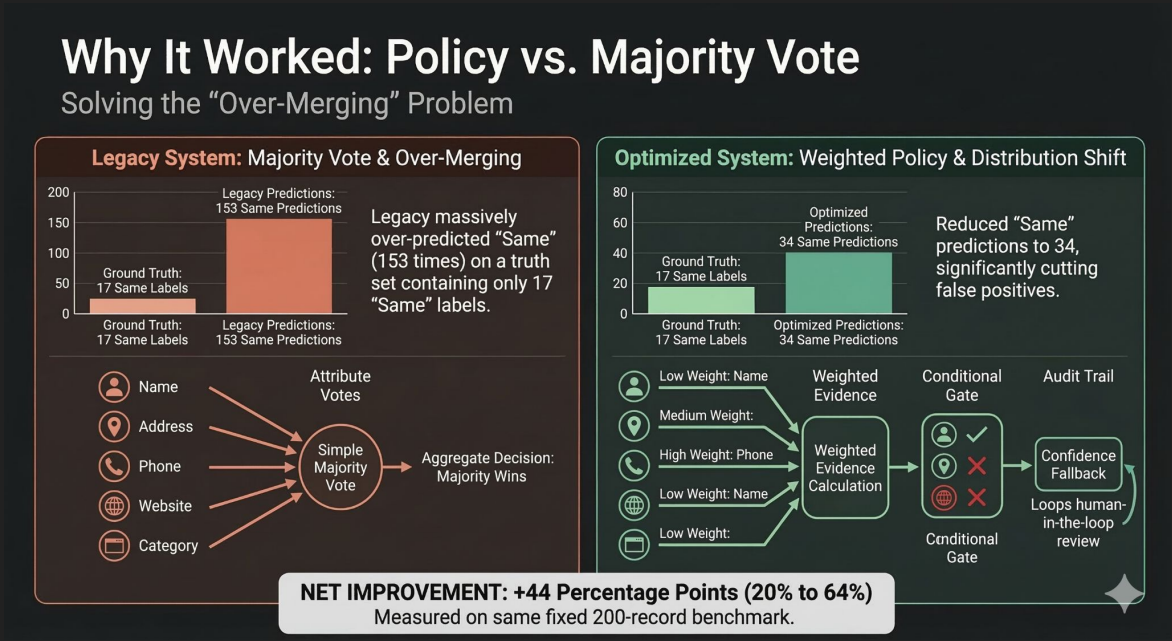
Attribute	F_1 score	Coverage
category	.8338	100%
phone	.8293	100%
website	.7305	100%
address	.2515	100%
name	.0408	100%

Record-Level Performance

The legacy and optimized ML models were compared in how they performed against a “golden 200” dataset to standardize the testing.

The mayhem model baseline scored 20% in accuracy whilst the optimized two layered system scored a 64% in accuracy.

Which is a +220% relative improvement.



Pipeline Integrity

There’s full data engineering pipeline to ensure the model’s data processing and training steps are logged, versioned, and repeatable

- Rigorous data splits 695 train / 155 val / 150 test with zero task overlap or leakage
- Adjudication workflow integrated labeling and adjudication to create the “golden” record sets.
- Automated eval evaluation is kept strictly separate from internal consistency scoring to avoid bias.

Note: This pipeline exists to create locked splits for experimentation and evaluation on the finalized attribute-level labels. Golden 200 is a benchmark used to evaluate record-level decision policies on the same 200 reviewed records.

How does this help overture?

The approach pushes towards building a production-ready framework for record conflation:

- Decisions are no longer a “black box"; we can see exactly which attribute weights triggered a merge.
- Every decision is reproducible via the locked data splits and versioned policy resolvers.
- the model and testing is also relatively lightweight

Wrap up

If given more time:

Try different ML/AI models:

- perceptrons?
- decision trees?
- Q-learning models?

Takeaways:

A bit more testing and review, and the data pipeline + scoring should be a great way of measuring any strategy/method of constructing a golden dataset.

Kate's Project

Expanded On Previous Work:

- Github: Mayhem_Attribute_Conflation (Authors - Jaskaran & Varnit)

Their Approach:

- Used rule-based baselines: Most Recent, Completeness, Confidence, Hybrid Heuristic
- Per-attribute ML trained on synthetic data

Attribute	Most Recent F1	Completeness F1	Confidence F1	Heuristic Hybrid F1	Starting ML F1
Address	0.8338	0.8338	0.7473	0.8338	0.2515
Category	0.8338	0.7031	0.7564	0.8094	0.6417
Name	0.8338	0.7319	0.7692	0.7667	0.3656
Phone	0.8554	0.8554	0.7591	0.8554	0.6929
Website	0.8323	0.8323	0.7574	0.8323	0.3483
Macro Avg	**0.8370**	**0.7913**	**0.7579**	**0.8195**	**0.4600**

My Approach

Strategy & Improvements

- Main goal: make ML more trustworthy, not just more complex
- Areas I improved:
  - Synthetic data realism using Yelp
  - Size of the labeled benchmark: `200 → 400`
  - ML confidence quality and decision thresholds
  - The interaction between ML and rule-based methods

SHIFT IN STRATEGY

Moving away from `ML vs rules`

Towards `ML selectively combined with rules`

Why Yelp?

- Provides records with same fields as our task
- Real labels were too limited for ML-only training
- Made it possible to generate many realistic training pairs cheaply

Why 200 More Labels?

- More edge cases covered
- More stable evaluation
- Increases confidence in our final performance conclusions

ML Improvements

Key Methodological Improvements

- **Per-attribute models** instead of one global model
- Improved synthetic generation:
  - Attribute-specific corruption
  - Near-equal and both-noisy cases
  - Labels based on quality, not confidence shortcuts

Calibration & Threshold Tuning

- Added **calibration-aware confidence** (Platt scaling)
- Tuned decision thresholds for F1 instead of fixed `0.5`
- **Why:**
  - Raw probabilities are often overconfident
  - Calibration aligns confidence with actual correctness
  - Different attributes naturally need different cutoffs
  - Optimized directly for validation F1

Milestone Performance Evolution (F1 Score)

Milestone	Address	Category	Name	Phone	Website	ML Macro F1
Starting ML	0.2515	0.6417	0.3656	0.6929	0.3483	0.4600
Early cleanup	0.7921	0.8338	0.2410	0.8000	0.8338	0.7001
Gated ML phase	0.7921	0.4623	0.8225	0.8519	0.7491	0.7356
Final ML refresh	**0.8315**	**0.8563**	**0.8079**	**0.8094**	**0.8563**	**0.8323**

Hybrid Model

Original Mayhem Hybrid

- Heuristic combination of rule signals
- Fully rule-based
- No learned routing between ML and rules

My Hybrid Idea

- Baseline is the safe default
- ML only overrides when confidence is strong enough
- Routing policy differs by attribute
- Searched `541,696` policy configurations

Example Decision Flow

- Baseline says: `base`
- ML says: `current`
- Calibrated ML confidence: `0.91`
- Name policy threshold: `0.85`

Final choice: ML
(Confidence > Threshold)

If confidence were 0.68 → Fall back to baseline
(Confidence < Threshold)

Final Results

- **Best final policy configuration:**
  - `name` : confidence-gated ML over `Most Recent`
  - `address` : confidence-gated ML over `Hybrid`
  - `phone` , `website` , `category` : baseline-only

Method Comparison (Macro F1)

Method	Macro F1
Starting ML	0.4600
Final ML	0.8323
Best baseline (`Most Recent`)	0.8574
Best swept hybrid	0.8491

Future Improvements—Learned Router

Idea:

- ❑ Learn when to trust ML vs Baseline
- ❑ Meta-model for routing decisions
- ❑ Could replace fixed threshold rules

Why is it promising:

- ❑ Can learn patterns fixed policies miss
- ❑ Can adapt differently by attribute or edge cases
- ❑ Could become better with more real labeled data

Reva’s Approach Summary

Three considerations:

- Whether the data for the place is objectively accurate
- Which source has better readability/more information
- Which source is better for each attribute, which source is better overall
- Classifications: alt, base, both, none (4); alt, base, both (3); alt vs base (2)

For determining validity of data

- HTTP GET requests and `phonenumbers` library to determine validity of websites and phone numbers
- Beautiful Soup and Google Maps API to cross-reference data

To predict which data has better readability/more information

- Rulebased and ML algorithms such as XGBoost, Random Forest, and SLMs

Future work turns scripts into a pipeline: validate data accuracy, 4-classification, if “none” then collects new data, final output is a complete and updated record

Feature Engineering

- Similarity Scores: uses `RapidFuzz` to calculate exactly how similar the names and addresses are (e.g., 95% similar).
- Completeness: It checks which source has *more* info
- Validation:
  - Web: HTTP GET requests to validate website existence (200 OK)
  - Phone: It uses the `phonenumbers` library to check if the number is valid in that country.
- Recency: It checks which source was updated most recently.

Dataset

- Golden Dataset (200 rows) - with a script to auto validate phone numbers and websites; 4 labels recalculated into 3 or 2 labels as needed
- Heuristic Weak Labels: For the other 1,800 rows, rule based and SLM classified data to provide the initial training signal
  a. Synthetic datasets to add more cases and balance dataset
- BeautifulSoup + web search and Google Places API for external validation & similarity calculations

Per-Attribute Similarity

Mean value similarity (0–100) vs golden per attribute.

Method	phone	web	address	category	n
Google API	13.0	5.0	82.81	0.0	200
Scrape + search	92.5	0.5	97.05	0.0	200

Per-Attribute Winner Agreement

Method	agreement_pct	n
Google API	21.5	200
Scrape + search	53.0	200

Per-Attribute Accuracy

Accuracy per attribute (method winner vs golden `attr_*_winner`) for methods with per-attribute outputs.

Method	phone_acc	web_acc	address_acc	category_acc	n
Google API	9.5%	16.5%	22.5%	37.5%	200
Scrape + search	27.0%	59.5%	68.5%	57.0%	200
SLM (Gemma3 4B)	20.0%	69.0%	40.0%	51.5%	200
SLM (Kimi)	45.5%	57.5%	70.5%	71.5%	200
SLM (GPT-4o mini)	34.5%	48.0%	68.5%	65.5%	200
XGBoost per-attr ensemble	11.5%	70.5%	45.5%	51.5%	200

4 class classification (base, alt, both, none)

Synthetic data generated (random rows paired for more “none” labels, attribute info removed to add more base or alt labels)

Method	Variant	accuracy	f1_macro	n	f1_none	f1_alt	f1_base	f1_both
SLM (Gemma3 4B)	recalculated	0.436170	0.200414	376	0.000000	0.000000	0.189655	0.612000
SLM (Kimi)	recalculated	0.433511	0.239250	376	0.000000	0.160000	0.185185	0.611814
SLM (GPT-4o mini)	recalculated	0.404255	0.247252	376	0.000000	0.282609	0.136986	0.569412
XGBoost 4-class	raw	0.380319	0.236168	376	0.000000	0.198347	0.188406	0.557920
XGBoost per-attr ensemble (4-class)	raw	0.406915	0.186839	376	0.027778	0.000000	0.126984	0.592593
SLM (Gemma3 4B)	raw	0.132979	0.069444	376	0.000000	0.000000	0.277778	0.000000
SLM (Kimi)	raw	0.151596	0.127147	376	0.000000	0.272300	0.236287	0.000000

3 class classification (base, alt, both)

SLM: perform zero-shot labeling of attribute winners & outputs final winner

- Recalculated labels based on individual attribute picks (detects “both” labels better)

Used Ollama models (kimi2 instruct, gpt4o-mini)

Confidence baseline rule based

- Strategy: based on Overture confidence scores. If `confidence > base_confidence`, pick `alt`.
- Heuristic: Includes a "Both-ness" detector where it defaults to both if similarities are extremely high and confidence scores are near identical.
- Rule-based limitation: can be easily swayed to wherever the dataset’s majority labels are

Method	Variant	accuracy	f1_macro	n	f1_alt	f1_both	f1_base
XGBoost per-attr ensemble	raw	0.505	0.277435	200	0.000000	0.676056	0.156250
Phase5 confidence baseline	raw	0.620	0.391008	200	0.293333	0.784452	0.095238
XGBoost 3-class	raw	0.585	0.265881	200	0.000000	0.743590	0.054054
SLM (Kimi)	raw	0.235	0.222808	200	0.343750	0.000000	0.324675
SLM Kimi recalc	recalculate	0.570	0.399120	200	0.225352	0.730627	0.241379

2-class classification (base, alt)

XGBoost: L2-Logistic Regression and a Gradient Boosted Classifier (GBC)

Method	Variant	accuracy	f1
XGBoost binary	raw	0.430	0.601399
Random Forest binary	raw	0.560	0.102041
XGBoost per-attr ensemble	raw	0.565	0.043956
SLM (Gemma3 4B)	raw	0.570	0.000000
SLM (Kimi)	raw	0.655	0.576687
SLM (GPT-4o mini)	raw	0.570	0.000000

Metric	Recorded "Recalculated" Stats
3-Class Accuracy	**77.5%**
Binary Accuracy	**81.5%**

Results

- how number of labels and labeling method affect accuracy
- Per-record labelling is much weaker than per-attribute labeling
- Rule based approaches can be limited by how the training data leans
- More balanced training datasets yielded better results for all XGBoost models
- Kimi with a rule based script to take it from 3 class to 2 class had the best results (81.5% binary accuracy)
- Future work:
  - Validation to classification pipeline
  - LoRA fine tuning

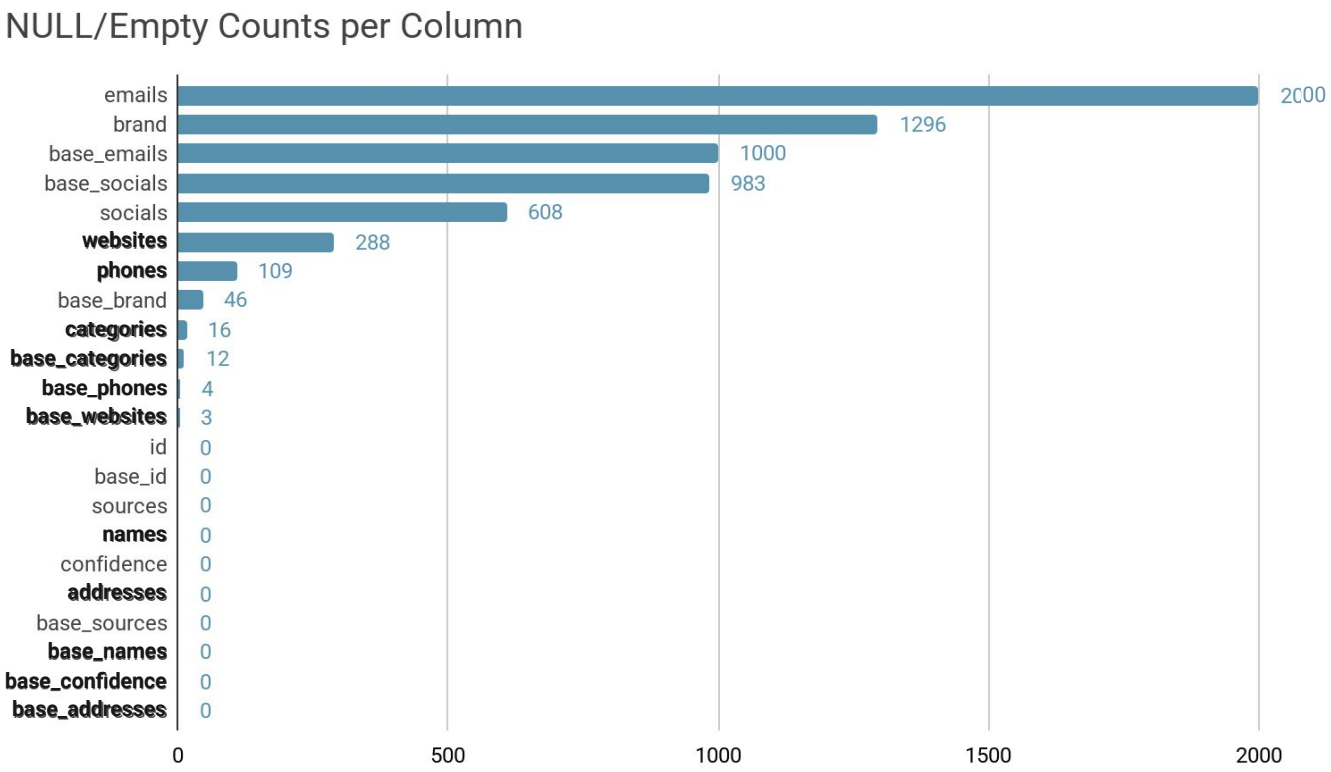
Neha's Approach

Initial Approach:

Rule Based

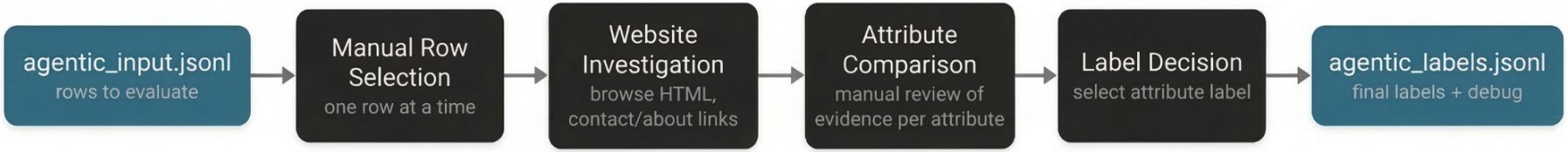
Findings:

Rule-based methods prioritize data structure, not data truth.

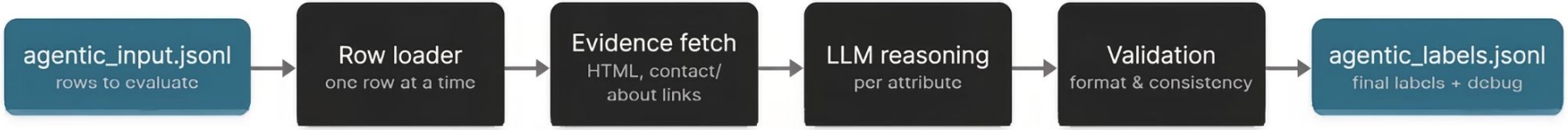


Approach 1 → Approach 2 → Results → Future

Manual Labeling



Technical Workflow



Findings

1. LLM is specific. Sometimes it's a good thing, and sometimes it’s not.

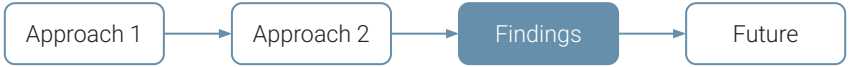
Location Name	Base Category	Alt Category	LLM Decision
Valley Transmission	automotive_repair	transmission_repair	transmission_repair
Mazda Nelspruit	used_car_dealership	car_dealership	used_car_dealership

Findings

2. What matters more? The primary category being better or holistic description being accurate?

Location: Ludwigs Biergarten (German burger restaurant)

	Primary	Alternate
Base	burger_restaurant	cafe, pizza_restaurant, salad_bar, food, restaurant
Other	restaurant	german_restaurant, burger_restaurant



Future Goals

1. Identify stronger LLMs with multilingual capabilities
2. Working with OMF to develop an outline for “better” category
3. Using the output to analyze which sources are best for which kinds of locations

Karthik's Approach

Idea:

I went for a rules based approach, defining specific rules to determine which of the pair is 'golden', based on format.

Through experimentation, the rules I devised were:

Attribute	Key Rules Used for Selection
Names	Normalize case and punctuation. If normalized names match → **B (Both)**. If names share ≥80% token similarity → **B**. Otherwise prefer names that are more descriptive (more words / reasonable length).
Categories	Compare the **primary category**. If both match → **B**. Prefer non-empty categories and slightly prefer more specific categories (more hierarchical tokens).
Websites	Normalize URLs (remove protocol, `www`, trailing slashes). If canonical domains match → **B**. Prefer **HTTPS**, penalize URL shorteners (e.g., `bit.ly`), penalize social domains when evaluating websites, prefer cleaner/shorter URLs.
Socials	Normalize URLs and compare canonical domains. If domains match → **B**. Penalize shortened links but allow social domains.
Emails	Prefer valid email format. Prefer **business domain emails** over free providers (e.g., Gmail, Yahoo). If identical emails → **B**.
Phones	Normalize phone numbers by removing punctuation. If digits match → **B**. Prefer numbers starting with **+ (E.164 format)** and numbers with valid length (10–15 digits).
Brand	Extract brand name from the structured field. If both empty → **N (Neither)**. If only one exists → choose that side. If normalized brand names match → **B**.
Addresses	Compare normalized address strings. If normalized addresses match → **B**. Prefer addresses with **more complete components** (street, locality, region, postcode, country). Slight preference for longer postal codes (e.g., ZIP+4).
Tie Breaking	If two scores are very close, use **source reliability** + **confidence score** to decide the winner.

Additionally I manually verified 200 labels through cross checking with the website, google, and yelp data to serve as comparison.

Karthik’s Approach

Findings:

Comparing my 200 manually labeled dataset to the automatically labelled dataset following the rules I coded, I ended with an F1 score of 0.71 with an accuracy of about 0.76.

```
attr        accuracy   macro_f1
names         0.760      0.484
categories    0.560      0.381
websites      0.550      0.453
socials       0.900      0.745
emails        1.000      1.000
phones        0.830      0.604
brand         0.980      0.982
addresses     0.540      0.395

Overall:
  accuracy = 0.765
  macro_f1 = 0.710
```

Challenges:

A surprising number of data values were in non-English language, making it difficult to understand both manually and through code.

Different languages have different address formats!

```
Per-attribute scores (final rules):
attr        accuracy   macro_f1
names         0.760      0.484
categories    0.560      0.381
websites      0.550      0.453
socials       0.900      0.745
phones        0.830      0.604
brand         0.980      0.982
addresses     0.540      0.395

Overall:
  accuracy = 0.731
  macro_f1 = 0.692
```

(without emails)

Takeaways:

Rule-based heuristics can effectively resolve many attribute conflicts.

Normalization is critical, structured attributes were easier to evaluate than semantic ones.

Categories and brand information require subjective judgment.

Data quality varied significantly across sources, highlighting the importance of source reliability and confidence scores.

Manually labeled golden datasets are essential for evaluating conflation methods and measuring algorithm performance.

Karthik’s Approach: Future

In the future, I could expand on:

- ★ Manually labelling more pairs to evaluate accuracy more reliably
- ★ Using Natural Language Processing techniques to better identify attributes like name, category, brand
- ★ Adapting a manual labelled golden dataset to work with ML approaches like teammates

Mruthula's Approach

Idea:

- deep dive into golden data set
- note down observations + potential issues
- create rule set for labelling
- create manually labelled golden dataset
- implement rule selection using DSPy and Google Colab

Golden Record Pick Logic

Attribute	Scenario	Rule	Examples/Notes
Name	Single language	Pick it	Example: Valley Transmission
Name	Native script + English transliteration combined	Pick combined field	Example: 京都御所 Kyoto Imperial Palace
Name	Native script only	Pick combined if exists, else native	Improves accessibility
Name	Multiple identical names	Pick preferred source	Remove location suffixes
Address	One source has postal code	Pick source with postal code	
Address	Both have postal codes	Pick preferred dataset	
Address	Street completeness differs	Pick more complete address	Pick the one with most comps: address, locality, region, postcode
Address	Missing optional info	Pick first-in-list or preferred source	7643 Gate Pkwy Ste 104 > 7643 Gate Pkwy
Phone	Multiple formats	Pick E.164 formatted number	+19049989600 > 9049989600
Phone	Different numeric lengths	Pick longest numeric	
Phone	One formatted	Prefer formatted	
Category	Different categories	Prefer more broader category	shipping_center preferred over vehicle_shipping
Category	Both valid	Pick preferred dataset	
Website	Same domain	Pick canonical full URL	https://www.goinpostaljacksonville.com
Website	Multiple domains	Pick official domain	
Website	Social or redirect links	Avoid	
Email	Multiple emails	Pick most official	
Email	Personal vs corporate	Prefer corporate	Look for if the brand name is in the email address
Socials	Multiple accounts	Pick official account	
Socials	Fan pages	Avoid	

Mruthula’s Approach

Model (DSPy + Google Colab)

- imported DSPy into Colab and used Ollama/Llama 3.1 8b model
- imported raw data + golden data sheets into Colab
- created BestAttributeSelector and MultiAttributeSelector
    - Implement rules in code and have model use ChainOfThought reasoning to pick ideal source
- tried to evaluate field-by-field accuracy and overall accuracy

Findings:

- could have gone with rules-based approach instead of automation/ML
- got very low accuracy in names/categories, other attributes not at all accurate
- International locations, so addressing/name conventions are different
- Had to deal with situations that I hadn’t considered before

```
🚀 RUNNING INFERENCE ON 10 ROWS...
✅ Row 0 | Name Choice: `A` or `B`
✅ Row 1 | Name Choice: Automotive Repair
✅ Row 2 | Name Choice: Car Dealer
✅ Row 3 | Name Choice: Red Wing
✅ Row 4 | Name Choice: Automotiive Repair
✅ Row 5 | Name Choice: Newport Institute
✅ Row 6 | Name Choice: Comfort Inn Alexandria West - Landmark
✅ Row 7 | Name Choice: SushiCo | Sushi & Bubble Tea Ede
✅ Row 8 | Name Choice: doctor
✅ Row 9 | Name Choice: Pick 'A' or 'B'
```

3	3	Red Wing	shoe_store

Mruthula’s Approach

Challenges:

- manual labelling = time-consuming
- covering as many real-world test cases as possible (missing data, discrepancies)
- selecting which records to construct in golden dataset
- new to DSPy, so learning the ropes and trying to get model running

Takeaways:

- labelling more data would have been helpful for the model to learn conventions more
- having consistent formatting in golden dataset (dicts vs extracting values)
- hard to translate rules into code that model understood
- could have done more research on what model would have been more optimal

FusePlace

Rule-Based Baseline + Random Forest ML for Attribute Conflation

0.83	+8pt	2,000
ML F1 Score	Over Baseline	Place Pairs

Satvik Khanna | sakhanna@ucsc.edu | Project A — Overture Maps Foundation

Approach — Dual-Path Conflation Pipeline

2K Overture Pairs
Meta + MSFT

Feature Extraction
~50 signals

Rule-Based Path
Confidence + Quality

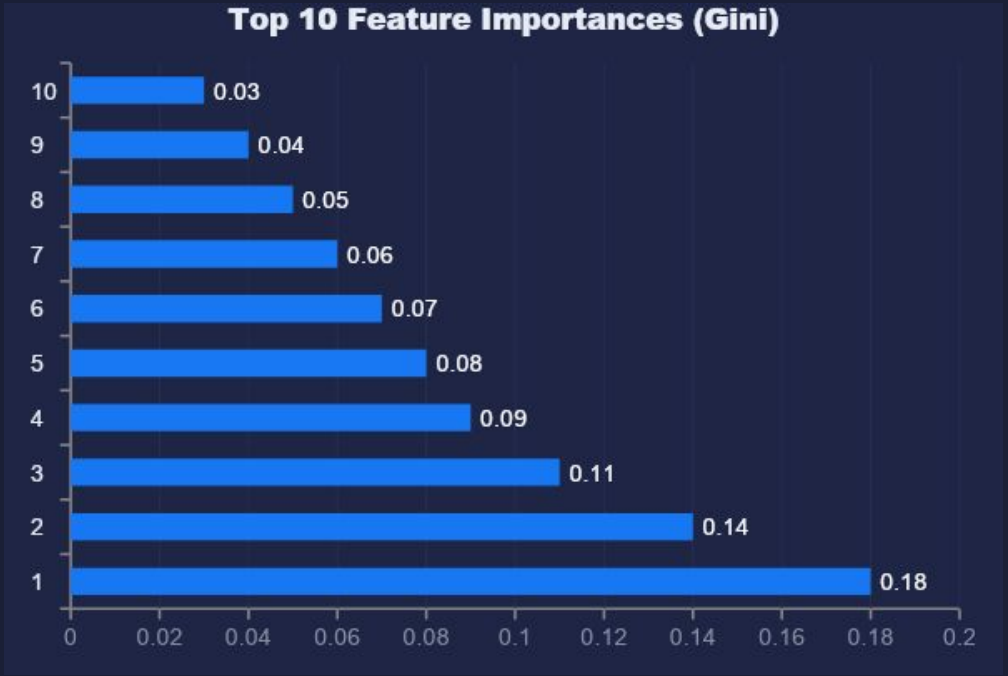
Random Forest Path
200 trees, d=10

Evaluate vs Golden
Stratified 80/20

Key Design Decisions

- Rules first — establish measurable baseline before adding complexity
- Single RF model shared across all 7 attributes — no per-attribute tuning needed
- No LLMs at inference — auditable, lightweight, reproducible
- Stratified 80/20 split, locked seeds — zero leakage
- **Single command: run_conflation**

Feature Engineering — 50+ Signals Per Pair



Similarity

RapidFuzz ratio, partial ratio, token sort ratio, edit distance, trigram overlap, Jaccard index

Confidence

Raw scores, delta, ratio, confidence × source count interaction

Quality

Validity flags, completeness %, token count, character length, null rate

Source

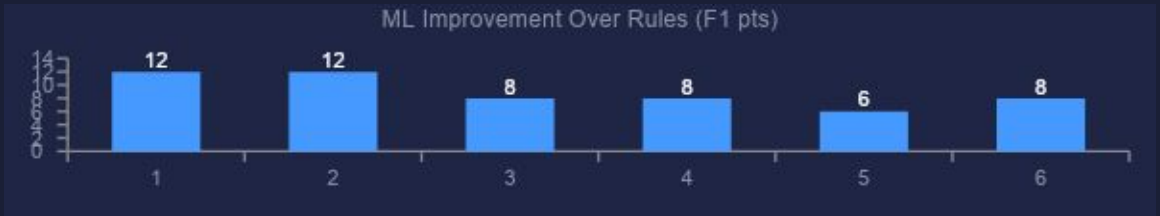
Source count, diversity, freshness timestamp delta

Attr-Specific

Phone: country code, digit count, E.164 flag. Web: HTTPS, domain match. Addr: field count, postal code

Model: Random Forest | 200 trees | max depth 10 | class-balanced | no hyperparameter search

Per-Attribute Performance Breakdown



Biggest Gains

Names: +12pt F1

Similarity features capture variants rules miss

Addresses: +12pt F1

Field-count + completeness signals help

Hardest Attribute

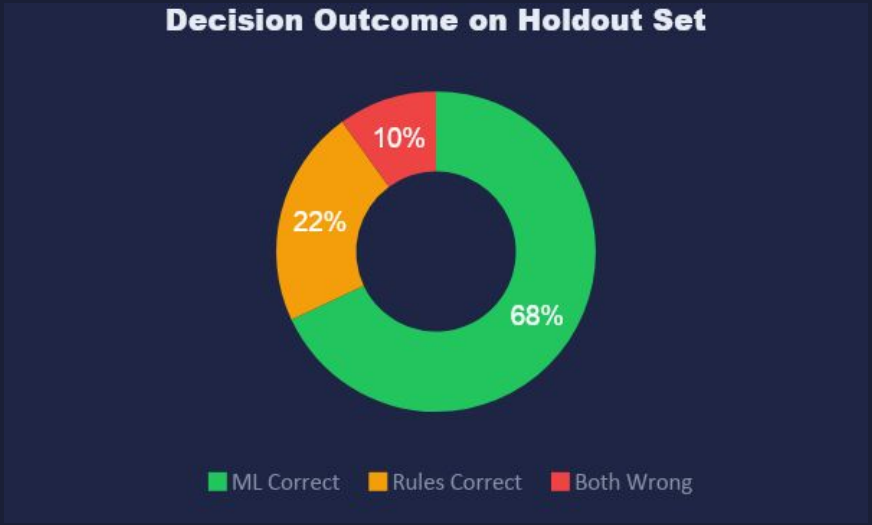
Websites: 0.66 F1

Yelp listing URLs vs actual business domains — both valid, semantically different

Needs domain-authority signal or web crawl validation

Aggregate Results — ML vs Rule-Based

Method	F1	Precision	Recall	Accuracy
ML (Random Forest)	**0.83**	0.87	0.79	0.81
Rule-Based Baseline	0.75	0.69	0.84	0.68



QnA